

# Introduction to Complex Numbers

Rail Engineering Mathematics

Sheffield Hallam University

Lecture 4

# Learning Outcomes

- Learn about the existence of  $j$ , the *imaginary number*.
- Recognise complex numbers.
- Add, subtract, multiply and divide pairs of complex numbers (in “Cartesian form”).

# Sets of Numbers

Numbers are understood to be organised in nested sets:

| Set                           | Symbol       | Examples                                                                 |
|-------------------------------|--------------|--------------------------------------------------------------------------|
| Natural numbers               | $\mathbb{N}$ | 0, 1, 2, 3, ...                                                          |
| Integers (i.e. whole numbers) | $\mathbb{Z}$ | $\mathbb{N}$ and -1, -2, -3, ...                                         |
| Rational numbers              | $\mathbb{Q}$ | $\mathbb{Z}$ and all fractions                                           |
| Real numbers                  | $\mathbb{R}$ | $\mathbb{Q}$ and all irrationals,<br>e.g. $\sqrt{2}$ , $\pi$ , $e$ , etc |

so

$$\mathbb{N} \subset \mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R}$$

Is  $\mathbb{R}$  the biggest set that exists, or are there still more numbers?

# Sets of Numbers

Quite often we are required to solve equations, such as

$$x - 7 = 0$$

Here the solution is 7, which is a member of  $\mathbb{N}$ .

We have also solved equations such as

$$x + 4 = 0,$$

but if do not accept the existence of any numbers other than the members of  $\mathbb{N}$ , then you cannot solve this equation.

If we define the solution as being  $x = -4$ , this idea leads to an entirely new set of numbers, the integers:  $\mathbb{Z}$ .

# Sets of Numbers

Once we have accepted the set  $\mathbb{Z}$ , we can solve a wider variety of equations. For example:

$$x + 12 = 0$$

$$(x + 9)(x + 2) = 0,$$

etc.

Therefore we benefit from accepting the existence of negative numbers such as -1.

# The Imaginary Unit

Up until this point, we assumed that there is no solution to the square root of a negative number. So, for example, there is no solution to the equation:

$$x^2 + 1 = 0$$

However, in a similar manner to the previous slides, if we can accept that another number set exists outside of  $\mathbb{N}$ ,  $\mathbb{Z}$ ,  $\mathbb{Q}$  and  $\mathbb{R}$ , then we can solve this equation.

Thus mathematicians define the solution to be the imaginary number  $j$ , so that:

$$j^2 + 1 = 0$$

# The Imaginary Unit

Let's examine the properties of our newly defined number.

If  $j^2 + 1 = 0$  then:

Definition of the imaginary unit:

$$j^2 = -1$$

and

$$j = \sqrt{-1}$$

We call  $j$  the **imaginary unit**, as it is the unit of the imaginary numbers in the same way that 1 is the unit of the real numbers.

# Imaginary Numbers

Hence, we can also express the square roots of the other negative numbers in terms of this unit  $j$ , for example:

$$\begin{aligned}\sqrt{-4} &= \sqrt{-1 \times 4} \\ &= \sqrt{-1} \times \sqrt{4} \\ &= j2\end{aligned}$$

and

$$\begin{aligned}\sqrt{-25} &= \sqrt{-1 \times 25} \\ &= \sqrt{-1} \times \sqrt{25} \\ &= \sqrt{j^2} \times \sqrt{25} \\ &= j5\end{aligned}$$

# Complex Numbers

Any multiples of  $j$ , such as  $j$ ,  $j6$ ,  $j0.3$ ,  $j\frac{7}{3}$  and  $j2\sqrt{3}$ , are all **imaginary numbers**.

When we combine these with real numbers through addition/subtraction, we create **complex numbers**, e.g.

$$6 - j5 \quad \text{and} \quad -3 + j$$

**Note:** Mathematicians (and most software) use  $i$ , while engineers use  $j$ . Thus, an engineer would write  $3 - j5$ , while a mathematician would write  $3 - 5i$ .

# Uses of Complex Numbers

Despite their apparent lack of physical meaning, complex numbers are essential for solving some real-world problems.

- Complex numbers can be used to represent certain engineering quantities, particularly in electronics:
  - AC currents and voltages
  - Impedances in AC circuits
- They are also used in:
  - The solution of differential equations
  - Aerodynamics (potential flow in two dimensions)
  - Control theory (stability analysis)
  - Signal processing (spectral analysis)

# Complex Numbers

The general (Cartesian/rectangular) form of a complex number is:

Standard Cartesian form of a complex number:

$$z = x + jy$$

where  $x$  and  $y$  are real numbers ( $\mathbb{R}$ ) and  $j = \sqrt{-1}$ .

We say that the **real part** of  $z$  is  $x$ :

$$\operatorname{Re}(z) = x$$

and the **imaginary part** of  $z$  is  $y$ :

$$\operatorname{Im}(z) = y \quad (\text{NOT } jy)$$

# Complex Number Arithmetic: Addition/Subtraction

To add/subtract complex numbers we simply add/subtract the corresponding real and imaginary parts separately.

$$\text{If } z_1 = 2 - j3, \quad z_2 = 6 - j2 \quad \text{and} \quad z_3 = -7 + j5,$$

calculate:

$$1) \quad z_1 + z_2$$

$$3) \quad z_2 - z_3$$

$$2) \quad z_2 - z_1$$

$$4) \quad z_1 + z_3$$

# Complex Number Arithmetic: Addition/Subtraction

## Solutions:

1)

$$\begin{aligned} z_1 + z_2 &= (2 - j3) + (6 - j2) \\ &= (2 + 6) + (-j3 - j2) \\ &= 8 - j5 \end{aligned}$$

2)

$$\begin{aligned} z_2 - z_1 &= (6 - j2) - (2 - j3) \\ &= 6 - j2 - 2 + j3 \\ &= (6 - 2) + (-j2 + j3) \\ &= 4 + j \end{aligned}$$

# Complex Number Arithmetic: Addition/Subtraction

## Solutions:

3)

$$\begin{aligned}z_2 - z_3 &= (6 - j2) - (-7 + j5) \\&= (6 - (-7)) + (-j2 - j5) \\&= 13 - j7\end{aligned}$$

4)

$$\begin{aligned}z_1 + z_3 &= (2 - j3) + (-7 + j5) \\&= (2 + (-7)) + (-j3 + j5) \\&= -5 + j2\end{aligned}$$

# Complex Number Arithmetic: Multiplication

To multiply complex numbers we simply expand the brackets; treating  $j$  just like any other constant.

Calculate:

1)  $z_1 z_2$

2)  $z_3 z_2$

# Complex Number Arithmetic: Multiplication

## Solutions:

1)

$$\begin{aligned}z_1 z_2 &= (2 - j3)(6 - j2) \\&= 12 - j4 - j18 + j^2 6 \quad (\text{remember } j^2 = -1) \\&= 12 - j22 + (-1)(6) \\&= 6 - j22\end{aligned}$$

2)

$$\begin{aligned}z_3 z_2 &= (-7 + j5)(6 - j2) \\&= -42 + j14 + j30 - j^2 10 \\&= -42 + j44 - (-1)(10) \\&= -32 + j44\end{aligned}$$

# Complex Number Arithmetic: Division

To divide complex numbers, we first make the denominator *real*.

This can be achieved by multiplying both the numerator and denominator by the **complex conjugate** of the denominator.

## Definition of complex conjugates:

For a complex number  $z = x + jy$ , the complex conjugate of  $z$  is:

$$\bar{z} = x - jy$$

Furthermore it can be proven that

$$z\bar{z} = x^2 + y^2$$

a result which is purely real and has no imaginary part.

# Complex Number Arithmetic: Division

Calculate:

$$1) \quad \frac{z_3}{z_2}$$

$$2) \quad \frac{z_1}{z_3}$$

# Complex Number Arithmetic: Division

## Solutions: 1)

$$\begin{aligned}\frac{z_3}{z_2} &= \frac{-7 + j5}{6 - j2} = \frac{(-7 + j5)(6 + j2)}{(6 - j2)(6 + j2)} \\&= \frac{-42 - j14 + j30 + j^210}{36 + j12 - j12 - j^24} \\&= \frac{-42 + j16 - 10}{36 + 4} \\&= \frac{-52 + j16}{40} = \frac{-13 + j4}{10} \\&= \frac{-13}{10} + j\frac{2}{5} = -1.3 + j0.4 \text{ to 2 d.p.}\end{aligned}$$

# Complex Number Arithmetic: Division

## Solutions: 2)

$$\begin{aligned}\frac{z_1}{z_3} &= \frac{2 - j3}{-7 + j5} = \frac{(2 - j3)(-7 - j5)}{(-7 + j5)(-7 - j5)} \\ &= \frac{-14 - j10 + j21 + j^2 15}{49 + j35 - j35 - j^2 25} \\ &= \frac{-14 + j11 - 15}{49 + 25} \\ &= \frac{-29 + j11}{74} \\ &= -\frac{29}{74} + j\frac{11}{74} = -0.39 + j0.15 \text{ to 2 d.p.}\end{aligned}$$